MOBILE APPLICATION DEVELOPMENT LABORATORY

Course Code	**22ISL71**	**CIE Marks**	**50**
L:T:P:S	**0:0:1:0**	**SEE Marks**	**50**
Hrs / Week	**3**	**Total Marks**	**100**
Credits	**01**	**Exam Hours**	**03**

Course outcomes:
At the end of the course, the student will be able to:

22ISL71.1	Create, test and debug Android application by setting up Android development environment.
22ISL71.2	Design User Interface and develop activity for Android App
22ISL71.3	Design and implement Database Application and Content providers
22ISL71.4	Implement adaptive, responsive user interfaces that work across a wide range of devices.

Mapping of Course Outcomes to Program Outcomes and Program Specific Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
22ISL71.1	3	3	3	2	3	-	1	1	-	1	-	1	3	3
22ISL71.2	3	3	3	2	3	-	1	1	-	1	-	1	3	3
22ISL71.3	3	3	3	2	3	-	1	1	-	1	-	1	3	3
22ISL71.4	3	3	3	2	3	-	2	1	-	1	-	1	3	3

Pgm. No.	List of Programs	Hours	COs
	Prerequisite Programs		
	No Prerequisite required		NA
	PART-A		
1.	Using Android SDK design and implement a single screen app that displays information about a small business.eg. Restaurant, Book shop etc. Your design must include: o Business name o Photo of business o Contact information	3	22ISL71.1

2.	Create an application to design a Visiting Card. The Visiting card should have a company logo at the top right corner. The company name should be displayed in Capital letters, aligned to the center. Information like the name of the employee, job title, phone number, address, email, fax and the website address is to be displayed. Insert a horizontal line between the job title and the phone number.	3	22ISL71.1
3.	Develop a simple application with one Edit text so that the user can write some text in it. Create a button called "Convert Text to Speech" that converts user input text into voice.	3	22ISL71.2
4.	Develop an application that makes use of the clipboard framework for copying and pasting of the text. The activity consists of two EditText controls and two Buttons to trigger the copy and paste functionality.	3	22ISL71.2
5.	Design an app for Tourist spot: With three activities, Welcome page, Display attractions of tourist spot and Webpage of the tourist spot.	3	22ISL71.2
6.	Develop an Android application using controls like Button, TextView, EditText for designing a Calculator having basic functionality like Addition, Subtraction, Multiplication and Division.	3	22ISL71.2
PART-B			
7.	Develop an android application Student Database App. The app should store USN, Student name and Semester of the student in SQLite database.	3	22ISL71.3
8.	Create an Android application with two screens where the user enters his information on the first screen. When a button is clicked, the entered data should be passed to the second screen and displayed there.	3	22ISL71.4
9	Develop an application to set an image as wallpaper. On click of a button, the wallpaper image should start to change randomly every 30 seconds.	3	22ISL71.4
10	Design and develop Health Monitoring App using Android. The app will store the blood pressure, blood group and glucose level of patient in SQLite database	3	22ISL71.3
11	Create a Sign Up activity where the user enters a username and password, with password validation based on the following rules: It must be at least 8 characters long, contain both uppercase and	3	22ISL71.4

	lowercase letters, include letters and numbers, and have at least one special character. On successful sign-up, use a Bundle to pass the username and password to the Login activity. In the Login activity, allow the user to sign in using the credentials created during sign-up. If the credentials match, navigate to a new activity displaying "Successful Login"; otherwise, show a toast message "Login Failed".		
12	Write a program to create an activity with two buttons START and STOP. On Pressing of the START button, the activity must start the counter by displaying the numbers from One and the counter must keep on counting until the STOP button is pressed. Display the counter value in a TextViewcontrol.	3	22ISL71.4

5)Develop an application that makes use of the clipboard framework for copying and pasting of the text. The activity consists of two EditText controls and two Buttons to trigger the copy and paste functionality.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayoutxmlns:android="http://schemas.android.com/apk/res/android"

android:layout_width="match_parent"

android:layout_height="match_parent"

android:orientation="vertical"

android:padding="20dp">


<EditText

android:id="@+id/editTextCopy"

android:layout_width="match_parent"

android:layout_height="wrap_content"

android:hint="Enter text to copy"

android:inputType="text"
```

```
android:layout_marginBottom="10dp" />

<Button
android:id="@+id/btnCopy"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Copy Text"
android:onClick="copyText"
android:layout_marginBottom="10dp" />

<EditText
android:id="@+id/editTextPaste"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Paste here"
android:inputType="text"
android:layout_marginBottom="10dp" />

<Button
android:id="@+id/btnPaste"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Paste Text"
android:onClick="pasteText" />
</LinearLayout>
```

MainActivity.java

```
package com.example.clipboarddemo;
```

```
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.content.ClipboardManager;
import android.content.ClipData;
import android.widget.EditText;
import android.view.View;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

EditTexteditTextCopy, editTextPaste;
ClipboardManagerclipboardManager;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

editTextCopy = findViewById(R.id.editTextCopy);
editTextPaste = findViewById(R.id.editTextPaste);
clipboardManager = (ClipboardManager) getSystemService(CLIPBOARD_SERVICE);
   }

   // User-defined method for Copy button
   public void copyText(View view) {
      String textToCopy = editTextCopy.getText().toString();
      if (!textToCopy.isEmpty()) {
```

```
ClipData clip = ClipData.newPlainText("copied_text", textToCopy);
clipboardManager.setPrimaryClip(clip);
Toast.makeText(this, "Text Copied!", Toast.LENGTH_SHORT).show();
        } else {
Toast.makeText(this, "Nothing to copy!", Toast.LENGTH_SHORT).show();
        }
    }

    // User-defined method for Paste button
    public void pasteText(View view) {
        if (clipboardManager.hasPrimaryClip()) {
ClipDataclipData = clipboardManager.getPrimaryClip();
ClipData.Item item = clipData.getItemAt(0);
            String pastedText = item.getText().toString();
editTextPaste.setText(pastedText);
Toast.makeText(this, "Text Pasted!", Toast.LENGTH_SHORT).show();
        } else {
Toast.makeText(this, "Clipboard is empty!", Toast.LENGTH_SHORT).show();
        }
    }
}
```

3)Develop a simple application with one Edit text so that the user can write some text in it. Create a button called “Convert Text to Speech” that converts user input text into voice.packagecom.example.texttospeech;

```
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.speech.tts.TextToSpeech;
```

```
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import java.util.Locale;

public class MainActivity extends AppCompatActivity {

EditTexteditText;
   Button btnSpeak;
TextToSpeechtts;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

editText = findViewById(R.id.editText);
btnSpeak = findViewById(R.id.btnSpeak);

      // Initialize TTS
tts = new TextToSpeech(this, new TextToSpeech.OnInitListener() {
         @Override
         public void onInit(int status) {
            if (status == TextToSpeech.SUCCESS) {
               int result = tts.setLanguage(Locale.getDefault()); // Use device language

               if (result == TextToSpeech.LANG_MISSING_DATA||
                     result == TextToSpeech.LANG_NOT_SUPPORTED) {
Toast.makeText(MainActivity.this, "Language not supported", Toast.LENGTH_SHORT).show();
               } else {
Toast.makeText(MainActivity.this, "TTS Ready", Toast.LENGTH_SHORT).show();
               }
            } else {
Toast.makeText(MainActivity.this, "TTS Initialization failed", Toast.LENGTH_SHORT).show();
            }
         }
      });
```

```
        // Speak on button click
btnSpeak.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String text = editText.getText().toString();
                if (text.isEmpty()) {
Toast.makeText(MainActivity.this, "Please enter text", Toast.LENGTH_SHORT).show();
                } else {
tts.speak(text, TextToSpeech.QUEUE_FLUSH, null, "tts1");

                }
            }
        });
    }

    @Override
    protected void onDestroy() {
        if (tts != null) {
tts.stop();
tts.shutdown();
        }
super.onDestroy();
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayoutxmlns:android="http://schemas.android.com/apk/res/android
"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:padding="20dp"
android:orientation="vertical"
android:gravity="center">

<EditText
android:id="@+id/editText"
android:layout_width="match_parent"
android:layout_height="wrap_content"
```

```
android:hint="Type something here"
android:inputType="text" />

<Button
android:id="@+id/btnSpeak"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Convert Text to Speech"
android:layout_marginTop="20dp" />

</LinearLayout>
```

4)Develop an application that makes use of the clipboard framework for copying and pasting of the text. The activity consists of two EditText controls and two Buttons to trigger the copy and paste functionality.

```
package com.example.program4_copy;



import android.content.ClipData;
import android.content.ClipboardManager;
import android.content.Context;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

EditTexteditText1, editText2;
    Button buttonCopy, buttonPaste;
ClipboardManagerclipboard;

@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

editText1 = findViewById(R.id.editText1);
editText2 = findViewById(R.id.editText2);
buttonCopy= findViewById(R.id.buttonCopy);
buttonPaste= findViewById(R.id.buttonPaste);

clipboard = (ClipboardManager) getSystemService(Context.CLIPBOARD_SERVICE);

buttonCopy.setOnClickListener(v -> {
            String text = editText1.getText().toString();
ClipData clip = ClipData.newPlainText("text", text);
clipboard.setPrimaryClip(clip);
        });
```

```
buttonPaste.setOnClickListener(v -> {
if (clipboard.hasPrimaryClip()) {
ClipData.Item item = clipboard.getPrimaryClip().getItemAt(0);
editText2.setText(item.getText());
            }
        });
    }
```

Program 6

Develop an Android application using controls like Button, TextView, EditText for designing a Calculator having basic functionality like Addition, Subtraction, Multiplication and Division.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:app="http://schemas.android.com/apk/res-auto"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:padding="20dp">


    <EditText

        android:id="@+id/etNum1"

        android:layout_width="0dp"

        android:layout_height="wrap_content"

        android:hint="Enter Number 1"

        android:inputType="numberDecimal"

        app:layout_constraintTop_toTopOf="parent"

        app:layout_constraintStart_toStartOf="parent"

        app:layout_constraintEnd_toEndOf="parent" />


    <EditText

        android:id="@+id/etNum2"

        android:layout_width="0dp"

        android:layout_height="wrap_content"
```

```
    android:hint="Enter Number 2"
    android:inputType="numberDecimal"
    app:layout_constraintTop_toBottomOf="@id/etNum1"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="16dp" />

<TextView
    android:id="@+id/tvResult"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:text="Result:"
    android:textSize="20sp"
    android:textStyle="bold"
    android:padding="10dp"
    app:layout_constraintTop_toBottomOf="@id/etNum2"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="20dp" />

<Button
    android:id="@+id/btnAdd"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:text="ADD"
    android:onClick="calculate"
    app:layout_constraintTop_toBottomOf="@id/tvResult"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toStartOf="@id/btnSub"
```

```
    android:layout_marginTop="20dp"
    android:layout_marginEnd="8dp"/>

<Button
    android:id="@+id/btnSub"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:text="SUB"
    android:onClick="calculate"
    app:layout_constraintTop_toBottomOf="@id/tvResult"
    app:layout_constraintStart_toEndOf="@id/btnAdd"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="20dp"
    android:layout_marginStart="8dp"/>

<Button
    android:id="@+id/btnMul"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:text="MUL"
    android:onClick="calculate"
    app:layout_constraintTop_toBottomOf="@id/btnAdd"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toStartOf="@id/btnDiv"
    android:layout_marginTop="20dp"
    android:layout_marginEnd="8dp"/>

<Button
    android:id="@+id/btnDiv"
```

```
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:text="DIV"
        android:onClick="calculate"
        app:layout_constraintTop_toBottomOf="@id/btnAdd"
        app:layout_constraintStart_toEndOf="@id/btnMul"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="20dp"
        android:layout_marginStart="8dp"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java package com.example.calculatorapp;

```
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    EditText etNum1, etNum2;
    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

```
    etNum1 = findViewById(R.id.etNum1);
    etNum2 = findViewById(R.id.etNum2);
    tvResult = findViewById(R.id.tvResult);
}

public void calculate(View view) {

    double n1 = Double.parseDouble(etNum1.getText().toString());
    double n2 = Double.parseDouble(etNum2.getText().toString());
    double result = 0;

    switch (view.getId()) {
        case R.id.btnAdd:
            result = n1 + n2;
            break;

        case R.id.btnSub:
            result = n1 - n2;
            break;

        case R.id.btnMul:
            result = n1 * n2;
            break;

        case R.id.btnDiv:
            if (n2 != 0)
                result = n1 / n2;
            else
                tvResult.setText("Result: Cannot divide by zero");
```

```
            return;
        }

        tvResult.setText("Result: " + result);
    }
}
```

Program 7

Develop an android application Student Database App. The app should store USN, Student name and Semester of the student in SQLite database.

MainActivity.Java

```
package com.example.prgm7;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

DBhelperdbHelper;
EditTextetUSN, etName, etSem;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
```

```
dbHelper = new DBhelper(this);

etUSN = findViewById(R.id.etUSN);
etName = findViewById(R.id.etName);
etSem = findViewById(R.id.etSem);
    }

    // Insert data into database
    public void insertStudent(View view) {
        String usn = etUSN.getText().toString();
        String name = etName.getText().toString();
        String semStr = etSem.getText().toString();

        if (usn.isEmpty() || name.isEmpty() || semStr.isEmpty()) {
Toast.makeText(this, "Please fill all fields", Toast.LENGTH_SHORT).show();
            return;
        }

        int sem = Integer.parseInt(semStr);
boolean inserted = dbHelper.insertStudent(usn, name, sem);

        if (inserted) {
Toast.makeText(this, "Inserted Successfully", Toast.LENGTH_SHORT).show();
etUSN.setText("");
etName.setText("");
etSem.setText("");
        } else {
Toast.makeText(this, "Insertion Failed (USN may already exist)",
Toast.LENGTH_SHORT).show();
```

```
        }
    }
}
```

DBhelper.java

```
package com.example.prgm7;

import android.content.Context;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import android.content.ContentValues;

public class DBhelper extends SQLiteOpenHelper {

    private static final String DATABASE_NAME = "StudentDB";
    private static final int DATABASE_VERSION = 1;
    private static final String TABLE_NAME = "students";
    private static final String COL_USN = "usn";
    private static final String COL_NAME = "name";
    private static final String COL_SEM = "semester";

    public DBhelper(Context context) {
        super(context, DATABASE_NAME, null, DATABASE_VERSION);
    }

    @Override
```

```
    public void onCreate(SQLiteDatabasedb) {
        String createTable = "CREATE TABLE " + TABLE_NAME + "("
                + COL_USN + " TEXT PRIMARY KEY, "
                + COL_NAME + " TEXT, "
                + COL_SEM + " INTEGER)";
db.execSQL(createTable);
    }

    @Override
    public void onUpgrade(SQLiteDatabasedb, int oldVersion, int newVersion) {
db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
onCreate(db);
    }

    // Insert method
    public booleaninsertStudent(String usn, String name, int sem) {
SQLiteDatabasedb = this.getWritableDatabase();
ContentValues cv = new ContentValues();
cv.put(COL_USN, usn);
cv.put(COL_NAME, name);
cv.put(COL_SEM, sem);
        long result = db.insert(TABLE_NAME, null, cv);
        return result != -1;
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:orientation="vertical"
android:padding="20dp"
android:layout_width="match_parent"
android:layout_height="match_parent">

<EditText
android:id="@+id/etUSN"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="USN" />

<EditText
android:id="@+id/etName"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Student Name" />

<EditText
android:id="@+id/etSem"
android:layout_width="match_parent"
android:layout_height="wrap_content"
```

```
android:hint="Semester"
android:inputType="number" />

<Button
android:id="@+id/btnInsert"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Insert"
android:onClick="insertStudent" />

</LinearLayout>
```

8)Create an Android application with two screens where the user enters his information on the first screen. When a button is clicked, the entered data should be passed to the second screen and displayed there

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayoutxmlns:android="http://schemas.android.com/apk/res/android"
android:orientation="vertical"
android:padding="20dp"
android:layout_width="match_parent"
android:layout_height="match_parent">

<EditText
android:id="@+id/etName"
android:hint="Enter your name"
android:layout_width="match_parent"
android:layout_height="wrap_content" />
```

```
<EditText
android:id="@+id/etAge"
android:hint="Enter your age"
android:inputType="number"
android:layout_width="match_parent"
android:layout_height="wrap_content" />

<Button
android:text="Submit"
android:onClick="sendData"     <!-- calls user-defined method -->
android:layout_width="wrap_content"
android:layout_height="wrap_content" />

</LinearLayout>
```

MainActivity.java

```
package com.example.twoscreenapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.content.Intent;
import android.view.View;
import android.widget.EditText;

public class MainActivity extends AppCompatActivity {

EditTextetName, etAge;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
```

```
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

etName = findViewById(R.id.etName);
etAge = findViewById(R.id.etAge);
    }

    // user-defined method called from XML
    public void sendData(View view) {
        String name = etName.getText().toString();
        String age = etAge.getText().toString();

        Intent intent = new Intent(this, SecondActivity.class);
intent.putExtra("userName", name);
intent.putExtra("userAge", age);

startActivity(intent);
    }
}
```

activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayoutxmlns:android="http://schemas.android.com/apk/res/android"
android:orientation="vertical"
android:padding="20dp"
android:layout_width="match_parent"
android:layout_height="match_parent">

<TextView
android:id="@+id/tvDisplay"
```

```
android:textSize="18sp"
android:layout_width="match_parent"
android:layout_height="wrap_content" />
</LinearLayout>
```

SecondActivity.java

```
package com.example.twoscreenapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.TextView;

public class SecondActivity extends AppCompatActivity {

TextViewtvDisplay;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_second);

tvDisplay = findViewById(R.id.tvDisplay);

        // Get data from Intent
        String name = getIntent().getStringExtra("userName");
        String age = getIntent().getStringExtra("userAge");

tvDisplay.setText("Name: " + name + "\nAge: " + age);
    }
}
```

9)Develop an application to set an image as wallpaper. On click of a button, the wallpaper image should start to change randomly every 30 seconds.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayoutxmlns:android="http://schemas.android.com/apk/res/android"
android:orientation="vertical"
android:gravity="center"
android:padding="20dp"
android:layout_width="match_parent"
android:layout_height="match_parent">

<Button
android:id="@+id/btnStart"
android:text="Start Changing Wallpaper"
android:onClick="startWallpaperChange"   <!-- user-defined method -->
android:layout_width="wrap_content"
android:layout_height="wrap_content" />
</LinearLayout>
```

MainActivity.java

```
package com.example.wallpaperchanger;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.app.WallpaperManager;
import android.os.Handler;
import android.view.View;
import java.io.IOException;
import java.util.Random;
```

```
public class MainActivity extends AppCompatActivity {


    int[] images = {R.drawable.img1, R.drawable.img2, R.drawable.img3,
R.drawable.img4};
    Handler handler = new Handler();
    Random random = new Random();
booleanisRunning = false;
    Runnable runnable;


    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
    }


    // user-defined method (called by onClick in XML)
    public void startWallpaperChange(View view) {
        if (isRunning) return; // prevent multiple starts
isRunning = true;


        runnable = new Runnable() {
            @Override
            public void run() {
changeWallpaperRandomly();
handler.postDelayed(this, 30000); // change every 30 seconds
            }
        };
handler.post(runnable); // start now
    }
```

```
    private void changeWallpaperRandomly() {
        int index = random.nextInt(images.length);
WallpaperManagerwallpaperManager = WallpaperManager.getInstance(this);
        try {
wallpaperManager.setResource(images[index]);
        } catch (IOException e) {
e.printStackTrace();
        }
    }

    @Override
    protected void onDestroy() {

super.onDestroy();
handler.removeCallbacks(runnable);
    }
}
```

Add Some Images

Add at least 3–4 images to your project:
🗁 **res/drawable/**

img1.jpg

img2.jpg

img3.jpg

img4.jpg

AndroidManifest.xml

```
<uses-permission android:name="android.permission.SET_WALLPAPER" />
```

11) Create a Sign Up activity where the user enters a username and password, with password validation based on the following rules: It must be at least 8 characters long, contain both uppercase and lowercase letters, include letters and numbers, and have at least one special

character. On successful sign-up, use a Bundle to pass the username and password to the Login activity. In the Login activity, allow the user to sign in using the credentials created during sign-up. If the credentials match, navigate to a new activity displaying "Successful Login"; otherwise, show a toast message "Login Failed".

Mainactivity.java

```
package com.example.program11;


import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.os.Bundle;
import android.view.View;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends
AppCompatActivity {

    EditText etUsername, etPassword;

    @Override
    protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        etUsername =
findViewById(R.id.etUsername);
        etPassword =
findViewById(R.id.etPassword);
    }

    // Method called from onClick in XML
    public void signUp(View view) {
        String username =
etUsername.getText().toString().trim();
        String password =
etPassword.getText().toString().trim();

        if (validatePassword(password)) {
```

```
            // Pass data using Bundle
            Bundle bundle = new Bundle();
            bundle.putString("username",
username);
            bundle.putString("password",
password);

            Intent intent = new Intent(this,
activity_login.class);
            intent.putExtras(bundle);
            startActivity(intent);
        } else {
            Toast.makeText(this, "Invalid
password! Follow rules.",
Toast.LENGTH_SHORT).show();
        }
    }


    private boolean validatePassword(String
password) {
        return password.length() >= 8 &&
                password.matches(".*[A-Z].*") &&
// at least one uppercase
                password.matches(".*[a-z].*") &&
// at least one lowercase
                password.matches(".*\\d.*") &&
// at least one digit

password.matches(".*[@#$%^&+=!].*"); // at least
one special char
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/
android"
    xmlns:app="http://schemas.android.com/apk/res-
auto"
```

```
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/etUsername"
        android:hint="Enter Username"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        app:layout_constraintTop_toTopOf="parent"

app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="150dp"/>

    <EditText
        android:id="@+id/etPassword"
        android:hint="Enter Password"
        android:inputType="textPassword"
        android:layout_width="0dp"
        android:layout_height="wrap_content"

app:layout_constraintTop_toBottomOf="@id/etUsernam
e"

app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="20dp"/>

    <Button
        android:id="@+id/btnSignUp"
        android:text="Sign Up"
        android:onClick="signUp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"

app:layout_constraintTop_toBottomOf="@id/etPasswor
d"

app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
```

```
        android:layout_marginTop="30dp"/>

</androidx.constraintlayout.widget.ConstraintLayou
t>
```

activity_login.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/
android"
    xmlns:app="http://schemas.android.com/apk/res-
auto"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/loginUsername"
        android:hint="Enter Username"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        app:layout_constraintTop_toTopOf="parent"
        android:layout_marginTop="150dp"

app:layout_constraintStart_toStartOf="parent"

app:layout_constraintEnd_toEndOf="parent"/>

    <EditText
        android:id="@+id/loginPassword"
        android:hint="Enter Password"
        android:inputType="textPassword"
        android:layout_width="0dp"
        android:layout_height="wrap_content"

app:layout_constraintTop_toBottomOf="@id/loginUser
name"

app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
```

```
        android:layout_marginTop="20dp"/>

    <Button
        android:id="@+id/btnLogin"
        android:text="Login"
        android:onClick="login"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"

app:layout_constraintTop_toBottomOf="@id/loginPass
word"

app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="30dp"/>

</androidx.constraintlayout.widget.ConstraintLayou
t>
```

activity_login.java

```
package com.example.program11;



import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.EditText;
import android.widget.Toast;

public class activity_login extends
AppCompatActivity {

    EditText loginUsername, loginPassword;
    String savedUsername, savedPassword;

    @Override
```

```
    protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);

        loginUsername =
findViewById(R.id.loginUsername);
        loginPassword =
findViewById(R.id.loginPassword);

        // Retrieve from bundle
        Bundle bundle = getIntent().getExtras();
        if (bundle != null) {
            savedUsername =
bundle.getString("username");
            savedPassword =
bundle.getString("password");
        }
    }

    public void login(View view) {
        String enteredUser =
loginUsername.getText().toString().trim();
        String enteredPass =
loginPassword.getText().toString().trim();

        if (enteredUser.equals(savedUsername) &&
enteredPass.equals(savedPassword)) {
            Intent intent = new Intent(this,
activity_success.class);
            startActivity(intent);
        } else {
            Toast.makeText(this, "Login Failed",
Toast.LENGTH_SHORT).show();
        }
    }
}
```

activity_success.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/
android"
    xmlns:app="http://schemas.android.com/apk/res-
auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">

    <TextView
        android:id="@+id/tvSuccess"
        android:text="Successful Login"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        app:layout_constraintTop_toTopOf="parent"

app:layout_constraintBottom_toBottomOf="parent"

app:layout_constraintStart_toStartOf="parent"

app:layout_constraintEnd_toEndOf="parent"/>

</androidx.constraintlayout.widget.ConstraintLayou
t>
```

activity_success.java

```
package com.example.program11;



import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class activity_success extends
AppCompatActivity {
    @Override
    protected void onCreate(Bundle
```

```
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_success);
    }
}
```

Program 12

Write a program to create an activity with two buttons START and STOP. On Pressing of the START button, the activity must start the counter by displaying the numbers from One and the counter must keep on counting until the STOP button is pressed. Display the counter value in a TextViewcontrol.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:app="http://schemas.android.com/apk/res-auto"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:padding="20dp">


    <TextView

        android:id="@+id/tvCounter"

        android:layout_width="0dp"

        android:layout_height="wrap_content"

        android:text="0"

        android:textSize="40sp"

        android:textStyle="bold"

        android:gravity="center"

        app:layout_constraintTop_toTopOf="parent"

        app:layout_constraintStart_toStartOf="parent"

        app:layout_constraintEnd_toEndOf="parent"

        android:layout_marginTop="40dp"/>
```

```
    <Button
        android:id="@+id/btnStart"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:text="START"
        android:onClick="startCounter"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toStartOf="@+id/btnStop"
        app:layout_constraintTop_toBottomOf="@id/tvCounter"
        android:layout_marginTop="40dp"
        android:layout_marginEnd="10dp"/>

    <Button
        android:id="@+id/btnStop"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:text="STOP"
        android:onClick="stopCounter"
        app:layout_constraintStart_toEndOf="@+id/btnStart"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintTop_toBottomOf="@id/tvCounter"
        android:layout_marginTop="40dp"
        android:layout_marginStart="10dp"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.counterapp;

import androidx.appcompat.app.AppCompatActivity;
```

```
import android.os.Bundle;
import android.os.Handler;
import android.view.View;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    TextView tvCounter;
    int count = 0;
    Handler handler = new Handler();
    boolean isRunning = false;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        tvCounter = findViewById(R.id.tvCounter);
    }

    // User-defined method for Start
    public void startCounter(View view) {
        if (!isRunning) {
            isRunning = true;
            handler.post(counterRunnable);
        }
    }

    // User-defined method for Stop
```

```
    public void stopCounter(View view) {
        isRunning = false;
        handler.removeCallbacks(counterRunnable);
    }

    // Runnable for continuous counting
    Runnable counterRunnable = new Runnable() {
        @Override
        public void run() {
            if (isRunning) {
                count++;
                tvCounter.setText(String.valueOf(count));
                handler.postDelayed(this, 1000); // count every 1 second
            }
        }
    };
}
```